

WWT Digital Architect Technical Challenge

Overview

This challenge aims to assess your competency in core skills required for the role and to provide a starting point for conversation during your technical interview. You are expected to spend at most 1-2 hours on this challenge. To ensure the integrity of the interview process, please don't share this challenge or the artifacts you produce publicly or privately with individuals that are not WWT employees.

Scenario

We have engaged with a (hypothetical) large pizza delivery chain to build out a hybrid web and mobile solution. The solution should be largely bespoke development, with a couple of key integrations noted below.

One complication is that in negotiation with the marketing department, it's been decided that dynamic content within the solutions will come from their existing headless CMS system (SeeEmLess) that provides RESTful and GraphQL access to content data. Access to media files is also provided. It is cloud based SaaS platform and provides an intuitive user interface to content producers. However, it's not designed to support high volume file access. In particular, there is a small cost for each endpoint read (including media) that has not been an issue for content creators yet.

The operations team is used to using PowerBI, but have been using a fairly manual procedure to ingest data. While they understand that long term they need to upgrade their data infrastructure, they are not currently interested in a big investment in a new data platform.

In addition, the team would like to use modern AI to streamline the user experience. An integrated chatbot should be included, which can answer questions related to menu items, order history and status, and add items to the user's cart. This can be text or voice activated.

Please provide an initial C4 model or other suitable architectural representation that provides a view of the following components of the system. Annotate your model to call out any key technical assumptions, and requirements, including NFRs

- Responsive web app
- Mobile app
- Administrative control app (for store and corporate users)
- Content integration
- Data integration
- Payment integration

The following high-level features and requirements should be represented

Feature	Consumer Apps	Store or Administrative control (desktop only)
User Registration and Login	✓	
Menu page	✓	
Cart	✓	
Store Selection	✓	
Checkout	✓	
Payment	✓	
Order history	✓	
Delivery updates/notifications	✓	
Chatbot	✓	
Notifications of new orders		✓
Update order status		✓
View Order history		✓

The expectation is that the service will be operating in one of the cloud providers, but the client is looking to you to provide an informed opinion (choose from AWS, Azure, or GCP).

Be aware that the brand has big plans and wants a product that will scale past 1000 locations. However, they haven't invested heavily in in-store infrastructure and are often plagued by network outages at their locations.